BME 423 Fall 2026: Homework 1

Due by 5 PM on Friday, September 25th, 2026. Please submit your work as a PDF file uploaded to Brightspace.

Please read the Homework Guidelines PDF before starting your assignment. Statistical tables necessary for completing this assignment are posted on Brightspace.

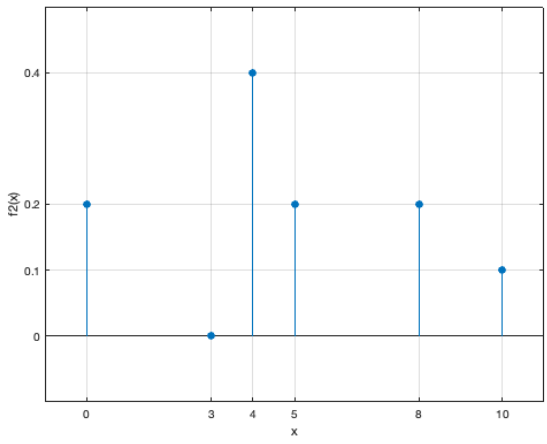
1. Classification of data (10 points)

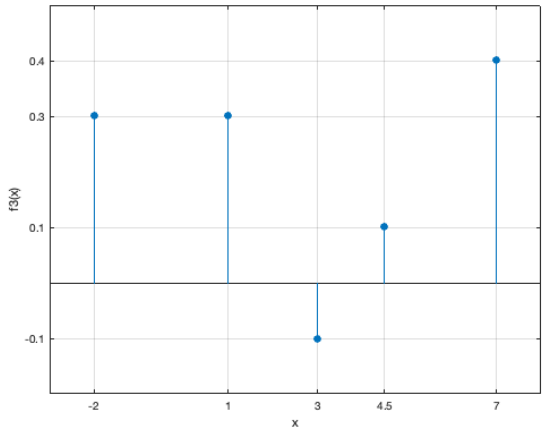
For each entry below, indicate whether the data are interval-continuous, interval-discrete, categorical-nominal, categorical-ordered.

a) Peak expiratory flow (L/min)
b) Country of birth
c) Age rounded to the nearest year
d) Answer to the question "How many text messages do you send per day?"
e) Systolic arterial blood pressure (mmHg)
f) Number of tender joints recorded as part of an arthritis diagnostic screen
g) Zip code
h) Resting transmembrane potential of a cell (mV)
i) Apgar score of a newborn infant
j) Height of a newborn baby (cm)

2. Discrete Random Variables (20 points)

(a) Which of the following functions could be examples of probability functions of a discrete random variable X? Explain your answers. List formulae for any properties of discrete RVs that you may wish to use and show complete work in testing those properties for each function.





f4(x) = 0.4 for x = .5
= 0.1 for x = 1
= 0.4 for x = 2
= 0 elsewhere

f5(x) = x/6 for x = 1, 2
= 1/2 for x = 3
= 0 elsewhere

(b) For the following probability function f6(x):

f6(x) = 0.55 for x = −1
= 0.15 for x = 2
= 0.1 for x = 3
= 0.2 for x = 7
= 0 elsewhere

- Sketch the probability function
- Sketch the distribution function $F_i(x)$ corresponding to each probability function
- Compute $E[X]$
- Compute $E[(X - E[X])^2]$
- Compute $E[X^2]$
- Compute $E[X^2] - (\mathrm{E}[\mathrm{X}])^2$

3. Continuous Random Variables (30 points)

(a) The probability density function (p.d.f.) for a continuous random variable X is given by:

$$f(x) = 0 \qquad x \leq 0$$
$$f(x) = 0.1 + x \qquad 0 < x \leq c$$
$$f(x) = 0 \qquad c < x$$

i. Find c and sketch the p.d.f.

ii. Find and sketch the distribution function (d.f.) of X

iii. Find
- $\mu = E[X]$
- $E[2X+1]$
- $P[X > 0.5]$
- $P[0 < X < 0.6]$
- $P[X = 0.5]$

[Provide by-hand, approximate sketches. Show work/calculations leading up to your final p.d.f. and d.f. sketches . If you choose, however, to use R to plot, please mention that and show your code and output.]

(b) Assume the fasting blood-glucose concentrations in healthy adults in the U.S. between 30 and 50 years of age is Normally distributed with a mean of 95 mg/dL and a standard deviation of 8.7 mg/dL (i.e., $X \sim N(95, 8.7^2)$ mg/dL).

i. Sketch and write the p.d.f. and d.f. of X
(Make sure that you show key features on the sketches of the p.d.f. and d.f. Showing the integral with proper terms and limits is enough for writing out the d.f.; you need not solve the integral.)

ii. Use the Standard Normal distribution (z table) in conjunction with the information given above to find/calculate the following:
- $\mu = E[X]$
- $\sigma^2 = E[(X - E[X])^2)$
- $E[X - 26]$
- 25th, 75th, and 97.5^{th} percentiles
- $P(X < 85)$
- $P(85 \leq X < 115)$
- $P(X \geq 115)$
- $P(X < 0)$
- $P(X = 95)$
- Check: Do the first three probabilities ($P(X < 85)$, $P(85 \leq X < 115)$, and $P(X \geq 115)$) add up to 1?

(c) Use R to confirm your answers for the following probabilities from Part (b)(ii):

- $P(X < 85)$
- $P(85 \leq X < 115)$
- $P(X \geq 115)$

Do your answers for Part (c) match the corresponding probabilities you calculated "by hand"? As part of your solution, provide your R code and the output generated by R.

4. Descriptive Statistics (20 points)

Seventy BME freshman students and 65 BME senior students constitute two different populations. Random samples of the heights of the students (in inches) have been drawn from each population (n = 10 in each sample group), and are given below:

Random sample from the freshman population
65 69 59 64 62 67 68 62 64 65
Random sample from the senior population
69 66 70 71 72 67 70 71 72 69

(i) Calculate the sample mean, median, variance, standard deviation, range, 75th percentile and standard error of the mean of each sample (i.e., calculate each statistic for the senior sample as well as for the freshman sample).

(ii) Estimate the population mean and population standard deviation for each of the two populations (seniors and freshmen).

(iii) How would you characterize the error in your estimate of the true population mean for each population?

(iv) How would this error change with changes in sample size?

(v) Does the senior population seem to be Normally distributed? Show how you came to this conclusion.

[Remember that for Normal distributions, fixed percentile indices correspond to percentile values of $\mu - \sigma$, $\mu + \sigma$, $\mu - 2\sigma$, etc. E.g., the 16th percentile value $\approx \mu - \sigma$. Use these properties to prove or disprove the Normal nature of this population based on the sample.]

(vi) Use R to construct q-q plots for each sample (i.e., one for freshman and one for seniors). Comment on any notable features of each plot. As part of your solution, provide your R code and the output generated by R.

(vii) Based on the sample mean and sample standard deviation of each sample, do you think that there is an actual difference in the heights of the senior and freshman populations? Provide a <u>qualitative</u> answer without performing any statistical tests.

5. Hypothesis Testing (20 points)

(a) Statistical inference using by-hand calculations
Xenograft tumor size data were collected for three groups comprised of 12 mice each, wherein every group was given a different dosing regimen of an isotype control antibody (an antibody of the same class/isotype as the antibody being studied, but that has no specificity for the epitope targeted by the actual antibody under investigation). Final tumor size measurements were taken 29 days after implantation of tumor cells, with group means and standard deviations presented in the table below.

Treatment Group	Mean (mm^3)	Standard Deviation (mm^3)
Group 1: 5 doses of 30μg	1188	189
Group 2: 5 doses of 100μg	1412	217
Group 3: Single dose of 200μg	1363	191

Determine if there is a difference among the three groups receiving different doses of the isotype control. Use a significance level of 0.05. (If you do find a difference, do not test which groups are different.)

(b) Statistical inference using R
B-type natriuretic peptide (BNP) levels give an indication of how healthy a subject's heart is (normal subjects have levels of BNP below roughly 100 pg/mL, while subjects suffering from various cardiac stressors and/or actual heart failure will have higher levels). A total of 165 subjects with similarly elevated BNP levels were randomly assigned to each of three different treatment groups, such that each group consisted of 55 subjects; one group acted as the control and received only a placebo injection, one group received an experimental drug, and one group received the standard treatment protocol. After receiving treatment, each subject's BNP levels were measured and stored for analysis.

These data are contained in the file BNP_data_HW1.csv available on the BME 423 course website, in the Homework 1 subfolder (Assignments → Homework 1). The first column contains the group identifier and the second column contains the actual measurements. Perform the appropriate statistical test using R to determine if there is a difference in BNP levels among the three different treatment groups based on the sample data in the file. Please provide your null hypothesis, the name of the statistical test being used, your R code (i.e., what you enter into R), the output generated by R, and your conclusion.